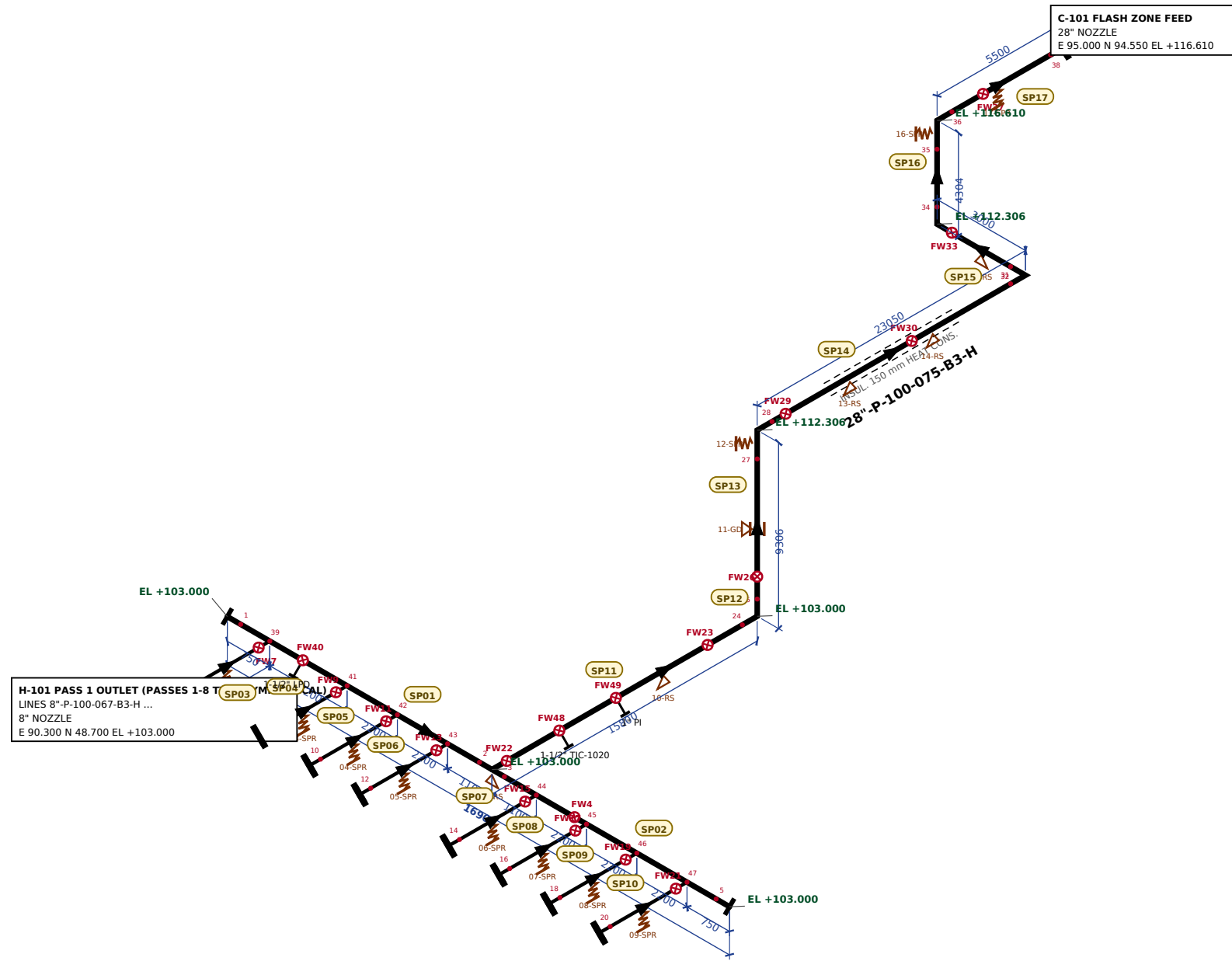
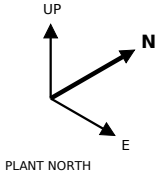


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



H-101 PASS 1 OUTLET (PASSES 1-8 T)
LINES 8"-P-100-067-B3-H ...
8" NOZZLE
E 90.300 N 48.700 EL +103.000

C-101 FLASH ZONE FEED
28" NOZZLE
E 95.000 N 94.550 EL +116.610

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	CAP, BW, WT 20.62, ASTM A234 WP9	28"	2
2	PIPE WT 20.62, BE, ASTM A691 9CR CL42 EFW	28"	64.8 m
3	TEE EQUAL, BW, WT 20.62, ASTM A234 WP9	28"	1
4	FLANGE WN RF CL300 B16.5, bore SCH 60, ASTM A182 F9	8"	8
5	PIPE SCH 60, BE, ASTM A335 P9 SMLS	8"	10.2 m
6	ELBOW 90 LR, BW, WT 20.62, ASTM A234 WP9	28"	5
7	FLANGE WN RF CL300 B16.47A, bore WT 20.62, ASTM A182 F9	28"	1
8	WELDOLET (branch), ASTM A182 F9	8"	8
9	SOCKOLET 3000# (branch), ASTM A182 F9	1-1/2"	2
10	SOCKOLET 3000# (branch), ASTM A182 F9	1"	1
11	ERECTION: GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	8"	8
12	STUD BOLT SET (nuts) A193 B16 / A194 4, CL300	8"	8
13	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	28"	1
14	STUD BOLT SET (nuts) A193 B16 / A194 4, CL300	28"	1
15	PIPE SUPPORTS (std. dwgs): guide x1, rest x5, spring x11	-	17

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	28"	7601	SP01-2	28"	2250	SP02-1	28"	5351
SP03-1	8"	1270	SP04-1	8"	1270	SP05-1	8"	1270
SP06-1	8"	1270	SP07-1	8"	1270	SP08-1	8"	1270
SP09-1	8"	1270	SP10-1	8"	1270	SP11-1	28"	11200
SP12-1	28"	3012	SP12-2	28"	1150	SP13-1	28"	6022
SP13-2	28"	1150	SP14-1	28"	11200	SP15-1	28"	8566
SP15-2	28"	866	SP16-1	28"	2171	SP16-2	28"	1150
SP17-1	28"	3083						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=11.2m SP02=5.7m SP03=1.4m SP04=1.4m SP05=1.4m SP06=1.4m SP07=1.4m SP08=1.4m SP09=1.4m SP10=1.4m
SP11=11.2m SP12=6.3m SP13=9.3m SP14=11.2m SP15=11.6m SP16=7.6m SP17=3.3m

NOTES

- Pipe/fittings/flanges per class B3 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 82.4 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Transfer line rises continuously from manifold to C-101 - no pockets (2-phase flow).
- Manifold symmetrical about C/L of heater: equal hydraulic length per pass.
- Rack crossing above tier 3 on dedicated transfer-line support beams at bents x=96/102.
- Formal stress analysis mandatory (API 560 terminal loads, C-101 nozzle WRC 537).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 28"-P-100-075-B3-H

H-101 OUTLET TRANSFER LINE TO C-101

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			
CFU-100-PI-ISO-001			
REV			
A			

LINE NUMBER	FROM	TO	PIPING CLASS
28"-P-100-075-B3-H	H-101 outlet manifold	C-101 flash zone	B3 - 300# 9Cr-1Mo ASTM A335 P9
DESIGN P / T	OPER. P / T	TEST P (HYDRO)	PIPE WALL
18.7 barg / 545.0 C	2.37 barg / 365.7 C	82.4 barg (1.5xPxST/S=2.94)	WT 20.62 (20.62 mm)
INSULATION	PWHT	NDE	STRESS
H 150 mm	YES - all thk (P-No.5B, B31.3 Tbl)	100 % RT of BW; 100 % MT/PT of fillets;	CAT 1 - T > 300 C & NPS >= 6; T > 200 C & NPS >= 1
P&ID	FLUID / PHASE	PAINT	SPOOLS / WELDS
CFU-100-PR-PID-004, CFU-100-PR-PID-006	P / M Atmospheric transfer lin	HT silicone/Al (insul.)	17 spools; 30 shop / 19 field